THE SECOND OPINION

The screener finds them. This does the homework.

A multi-agent equity research crew that produces one-page, fully cited due-diligence reports

Davon Crane · Building Agentic AI Systems — Final Project · August 2026

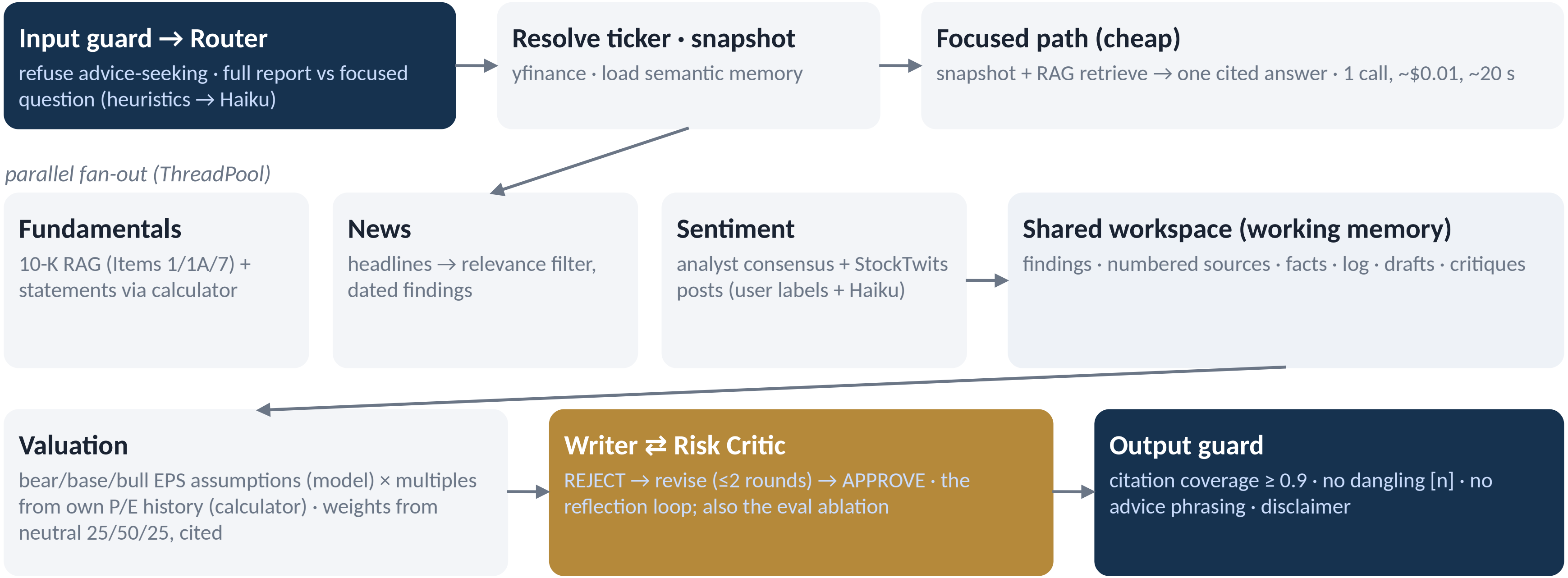
github.com/DavonCrane/second-opinion

The bar to beat: a Google search + one chatbot call

Everything needed to evaluate a company is public. Synthesizing it takes hours, and a single LLM prompt gives you a confident, uncited summary.

1. **Primary sources, cited by paragraph**
   RAG over the 10-K — “Item 1A ¶7”, not an article about it
2. **Sources cross-examined**
   Sentiment vs. fundamentals vs. filing language — divergence is called out
3. **A bear case as strong as the bull case**
   Held to the same evidentiary standard; a critic rejects weak ones
4. **Memory**
   “Since last analysis” — what changed since you last looked
5. **Code does the math**
   12-month view from visible, disputable assumptions — never an oracle number

Architecture — a deterministic orchestrator over six agents



→ report.md (+PDF) · episodic memory (this run) · semantic memory (durable facts) · Streamlit dashboard

Deliberately NOT a ReAct planner: the workflow has a known shape, so code-driven control is cheaper, testable offline, and produces an identical, observable trace every run.

Seven of eight course patterns — and one left out on purpose

1

Routing + parallelization

router.py; ThreadPool fan-out of 3 analysts

2

Reflection / self-critique

Risk Critic → writer revision loop (≤2 rounds)

3

Tool use — 6 tools

yfinance · SEC EDGAR · news · StockTwits · Reddit · calculator

4

Multi-agent with handoffs

6 agents over a shared workspace

5

Memory — 3 types

working · episodic · semantic

6

RAG with citations

10-K Items 1/1A/7 → ‘Item 1A ¶7’

7

Guardrails + eval harness

input/output/weight guards · 12-case eval, 3 arms

8

NOT: LLM planning (ReAct)

known workflow shape → deterministic orchestrator instead

Each pattern is exercised by a named test in tests/ — 27 tests run fully offline against fixtures and a scripted model.

The product: a one-page, fully cited report

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NVDA — NVIDIA Corporation

Generated: 2026-08-16 · **Price at analysis:** $225.16 · **Data:** 10-K 2026-01-25, market data & news to 2026-08-16

Profile: High-quality AI infrastructure grower, premium valuation, concentrated customer risk [1][16]

SNAPSHOT

METRIC	VALUE	SOURCE
Revenue growth (YoY)	+65.5% (prior yr +114.2%)	[1]
3-yr revenue CAGR	+100.0%	[1]
Gross / operating margin	71.1% / 60.4%	[1]
Rule of 40	**126** (65.5% + 60.4%)	[1]
Free cash flow	$96.7B (+58.9% y/y)	[1]
Total debt / net cash (cash + ST investments)	$11.0B / $51.5B	[1]
Debt-to-equity · interest coverage	0.07 · 547.1×	[1]
P/E (trailing / forward)	34.5× / 17.6×	[1]
Analyst consensus	Strong Buy 7.7/10 (n=61) · target $302.83 (+34%)	[2]
Retail sentiment	not available this run (analyst-only)	[2]

WHAT THE COMPANY DOES

NVIDIA operates primarily as a data-center-scale AI infrastructure company, organized into Compute & Networking (data center accelerated computing, AI platforms, automotive) and Graphics (GeForce, Quadro/RTX) segments [1][17]. Data Center revenue growth of 68% y/y is now outpacing both Gaming (+41%) and Professional Visualization (+70%), underscoring that AI infrastructure has become the dominant driver of the business mix [1][17].

FINANCIAL HEALTH

The balance sheet is very conservative, with $62.6B in cash and marketable securities against just $11.0B of total debt, yielding net cash of $51.5B, a debt/equity ratio of 0.07, and interest coverage of 547.1x [18]. Free cash flow reached $96.7B, up 58.9% y/y, broadly tracking revenue growth, and management states it believes liquidity is sufficient for at least the next twelve months [17][18].

BULL CASE

1. **Data Center-led growth engine** — Data Center revenue grew 68% y/y in fiscal 2026, outpacing Gaming (+41%) and driving overall revenue to $215,938M, up 65.5% y/y [1][17].
2. **Strong Street conviction** — Wall Street consensus is strong_buy at 7.7/10 bullishness across 61 analysts, with a mean 12-month target of $302.83, implying +34.5% upside from the current price [2].
3. **Expanding footprint into power infrastructure** — NVIDIA plans to invest up to $3 billion in Lancium, a power infrastructure developer, extending its reach into the energy supply chain underpinning AI data centers, while hedge funds have simultaneously increased stakes in NVIDIA's key foundry supplier, Taiwan Semiconductor [8][4].

BEAR CASE

1. **Customer concentration risk** — The 10-K discloses that one direct customer accounted for 22% of total revenue and another for 14% in fiscal 2026, a specific and quantified dependency that could undercut growth durability if either customer's capex plans shift [16].
2. **Growth deceleration alongside margin compression** — Revenue growth slowed sharply to +65.5% y/y from +114.2% the prior year, while gross margin compressed 3.9 points to 71.1% from 75.0%, even as operating margin remains high at 60.4% [17].

Live NVDA report from the evaluation run (Aug 16, 2026)

- ### Snapshot
  growth, margins, Rule of 40, debt, P/E, analyst consensus, retail sentiment — every row sourced
- ### Bull case & bear case
  3 bullets each; the critic rejects a bear case vaguer than the bull case
- ### 12-month scenarios
  bear/base/bull EPS × own P/E bands → implied prices; Claude's weighted view beside the analyst mean
- ### Sentiment check
  cross-examines Street/retail mood against fundamentals — names the divergence
- ### Since last analysis
  episodic memory: price change, new findings, weight shifts
- ### Sources + disclaimer
  numbered, paragraph-level 10-K citations; 'not investment advice' enforced by the guard

Evaluation — 12 live cases, three arms

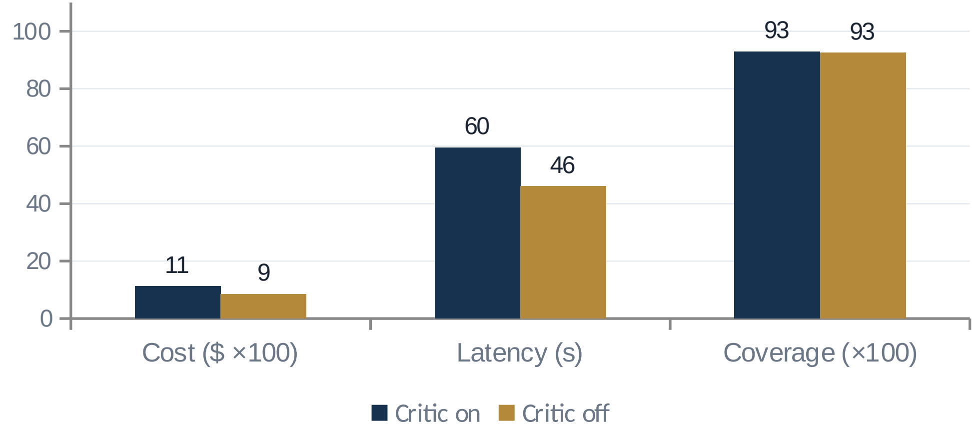
12 / 12
task success

0.936
mean citation coverage

$0.11
per full report

±3 pts
weight stability (5 runs, max σ 0.031)

Ablation: Risk Critic on vs. off (full reports)



Cases

- NVDA · MSFT · JPM (bank: Rule of 40 = n/a) · RDDT (recent IPO)
- RIVN — loss-making: scenario table correctly declined
- “nvidia” → name resolution · ZZZZ → clean error
- “NVDA — what’s their debt?” → cheap path, 1 call, cited
- “Should I buy NVDA?” → refused, $0
- re-run → since-last-analysis diff · news outage → degrades

Honest findings — what the eval actually taught me

The critic barely moves the metrics — and that's the point

Critic off: same 12/12, coverage −0.4 pt, 24% cheaper, 23% faster.

Why: during development the critic rejected drafts for real reasons — a bear bullet that said only “the stock is expensive”, a bear case vaguer than the bull case. Each fix moved into the writer’s prompt. The critic’s lessons migrated upstream.

What remains is insurance: it still rejected 2 of 5 stability-arm drafts with concrete issues. Catch rate went from ~4 in 10 before prompt hardening to ~2 in 14 after.

Decision: keep it on by default; expose --no-critic for cheap bulk screening.

1 RAG index contaminated by fixture data

Illustrative 10-K chunks shared IDs with the real filing; the model itself flagged ‘figures don’t reconcile’. Fix: filing-scoped chunk IDs + retrieval.

2 Guardrail false positive

“no guaranteed follow-through” tripped the advice filter and trapped the critic in two useless revisions. Fix: promissory-usage pattern + a test.

3 Model reversed a comparison

“price sits below all three scenarios” — it didn’t. Fix: that sentence is now generated by code. Principle: LLM chooses assumptions, code does arithmetic.

What's next, and where to find it

- **Dashboard**
  Streamlit UI: live agent log, critic verdicts, price & scenario map, follow-up questions, PDF export — cloud-deployable with a password gate

- **Future work**
  X/FinTwit as a third sentiment source · 10-Q ingestion · rubric-scored critic · self-hosted deployment

- **Limitations stated**
  P/E scenarios don’t apply to loss-makers (declined, not forced) · weights measured for stability, not validated as forecasts · yfinance/StockTwits coverage varies

Repository

github.com/DavonCrane/second-opinion

README — run it from scratch
docs/architecture.md
eval/eval_report.md
27 tests · pytest
tag: final-submission

Not investment advice. A research-synthesis tool that reads public sources and cites them.

Thank you — Davon Crane